

Assignment 1 : Implement multi-threaded client/server Process communication using RMI.

Terminal Set up :

1. Open terminal and check java version is 21 : `javac --version`
2. Run : `sudo update-alternatives --config java` → then select option of java of 21 version using 0,1 or 2 depends
3. Run : `sudo update-alternatives --config javac` → then select option of java of 21 version using 0,1 or 2 depends
4. Run : `sudo apt update`
5. Then follow below steps for program

6. Create Files

Create 4 Java files:

- Hello.java
- HelloImpl.java
- Server.java
- Client.java

7. Steps for terminal

Step 1: Open terminal and create new folder

```
cd Desktop
mkdir RMI
cd RMI
```

Step 2 : Create 4 files `touch Hello.java HelloImpl.java`

`Server.java Client.java`

Step 3: Compiles all files

Compile Java Files `javac *.java`

Step 4: **Run Server**

Open Terminal 1: `java Server`

Step 5: Run Client

Open Terminal 2: `cd`

Desktop/RMI `java Client`

1. Check Java Installation

Open terminal and type:

```
java --version and javac
```

```
--version
```

If Java is installed, you will see version output like:

```
openjdk 17
```

2. If Java is NOT Installed

Install Java JDK using:

```
sudo apt update
```

```
sudo apt install default-jdk -y
```

3. Verify Installation

```
java --version  
javac --version
```

🔗 First Check This and then create file

Viva Questions & Answers for RMI Assignment

1. What is Distributed System?

A distributed system is a collection of multiple computers that communicate and work together as a single system.

2. What is RMI in Java?

RMI stands for Remote Method Invocation.

It allows a Java program to call methods of an object located on another machine.

3. What is the use of RMI?

RMI is used for communication between client and server applications in distributed systems.

4. What are the main components of RMI?

- Remote Interface
- Remote Object (Implementation Class)
- Server
- Client
- RMI Registry

5. What is a Remote Interface?

A remote interface defines methods that can be called remotely by the client.

Example:

```
public interface Hello extends Remote
```

6. Why do we extend Remote interface?

To make methods accessible remotely.

7. Why is RemoteException used?

It handles communication/network-related errors during remote method calls.

8. What is UnicastRemoteObject?

It is a class used to export a remote object so that it can receive incoming remote calls.

9. What is RMI Registry?

RMI Registry stores remote objects with names so clients can find them.

Default port:

1099

10. What does rebind() do?

It binds or registers the remote object with the RMI registry.

Example:

```
reg.rebind("HelloService", obj);
```

11. What does lookup() do?

Client uses lookup() to search and access the remote object from registry.

12. Why multi-threading is used in this assignment?

Multi-threading allows multiple clients to communicate with the server simultaneously.

13. What is a Thread?

A thread is a lightweight process used for parallel execution.

14. Which class is used for multi-threading in your code?

class ClientThread extends Thread

15. What is the role of server in RMI?

The server creates and registers remote objects for clients.

16. What is the role of client in RMI?

The client accesses remote objects and invokes methods remotely.

17. Which protocol is used in RMI?

TCP/IP protocol is used internally.

18. Difference between RMI and Socket Programming?

RMI	Socket
High-level communication	Low-level communication
Supports remote methods	Supports data transfer only
Easier in Java	More coding required

19. Advantages of RMI

- Easy distributed communication
 - Object-oriented approach
 - Supports multi-threading
 - Secure and scalable
-

20. Disadvantages of RMI

- Java dependent
 - Slower than local calls
 - Requires network connectivity
-

21. What is the output of your assignment?

Server handles multiple client requests and returns messages like:

Hello Client-1 from Server

22. What happens if server is not running?

Client gets connection error because remote object is unavailable.

23. Why synchronized method is used? public synchronized String message()

It prevents multiple threads from accessing the method at the same time.

24. What is the difference between process and thread?

Process	Thread
Heavyweight	Lightweight

Separate memory Shared memory

Process	Thread
Slower	Faster

25. Real-life examples of Distributed Systems

- Google Search Engine
- Banking Systems
- Cloud Computing
- Online Games
- ATM Network